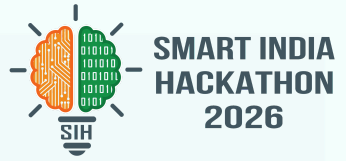


SMART INDIA HACKATHON 2026



Evidence Driven Dark Web Threat Actor Attribution

- **PROBLEM STATEMENT ID – SIH26151**
- **PROBLEM STATEMENT TITLE – DARK-WEB THREAT ACTOR DE-ANONYMIZATION**
- **THEME – BLOCKCHAIN & CYBERSECURITY**
- **PS CATEGORY – SOFTWARE**
- **TEAM ID – 125221**
- **TEAM NAME – TEAM PHANTOM**



Evidence Driven Dark Web Threat Actor Attribution

CONTEXT & PROBLEM

India recorded **1,01,928** cybercrime cases in **2024 (+19% YoY)**; **89.9%** of dark web chatter specifically targets Indian entities*. If **India** is such a major target, why does so much underground threat activity remain difficult to investigate & attribute?

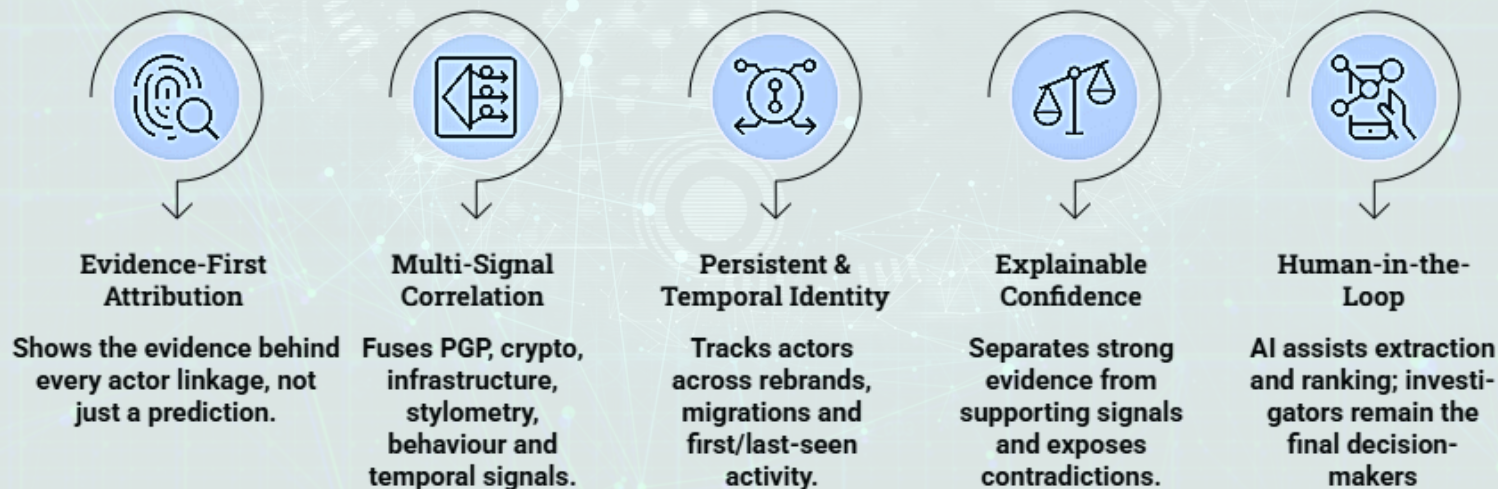
- Tor conceals server location, limiting direct IP attribution.
- Actors constantly change identities across platforms .
- Evidence remains fragmented across multiple sources .
- Wallet & stylometry signals weaken against modern evasion techniques.
- Dark web content is volatile, with services frequently changing or disappearing.

How can fragmented underground observations be correlated into an explainable persistent threat-actor attribution?

SOLUTION & HOW IT ADDRESSES THE PROBLEM

- **Multi source evidence collection**: gathers infrastructure, PGP, wallet, text and behavioural signals from permitted sources.
- **Persistent Identity Fingerprint (PIF)**: links recurring handles, templates, PGP/wallet clues and other fingerprints across rebrands and migrations.
- **Evidence Fusion Attribution**: combines independent signals using source reliability instead of relying on a single indicator.
- **Adversarial aware Stylometry**: detects AI-assisted/short text and shifts weight toward stronger behavioural and temporal evidence.
- **Temporal Actor Graph**: connects entities across marketplaces, forums and infrastructure using first seen/last seen observations.
- **Explainable Confidence Scoring**: shows exactly which evidence supports or contradicts an attribution hypothesis.

UNIQUENESS AND INNOVATION



GITHUB REPO



1. Dark-Web Collection

IMPLEMENTATION: Tor-aware Python crawlers → Playwright for dynamic pages → isolated circuits per domain → scheduled re-crawling

OUTPUT: Raw page + onion URL + timestamp + source + cryptographic hash.

2. Evidence & Entity Extraction

IMPLEMENTATION: Parse collected pages using rule-based + NLP extractors.

EXTRACT: Handles | PGP fingerprints | Wallets | Onion URLs | Domains | Infrastructure | Text | Timestamps. Normalize identifiers before correlation.

3. Persistent Identity Correlation

IMPLEMENTATION: Candidate generation via Exact PGP reuse, Wallet/address relationships, Infrastructure overlap, Handle similarity. Temporal checks including First seen / last seen, Activity overlap, Migration between services, Store relationships in a Neo4j actor graph.

4. Behaviour + Stylometry Analysis

IMPLEMENTATION: Text features: vocabulary, sentence length, punctuation, function-word patterns. Combine with: Posting frequency, Time-of-day patterns, Activity intervals, Recurring behavioural patterns

5. Cryptocurrency & Infrastructure Analysis

CRYPTO: Extract wallet addresses → normalize → transaction/relationship analysis → candidate clustering. **INFRASTRUCTURE:** Compare HTTP headers, Server responses, TLS/certificate information, Exposed status pages, Onion-service metadata. These generate candidate infrastructure relationships, not automatic attribution.

6. Evidence Fusion & Attribution

Build an evidence graph / factor graph. Each signal receives: Evidence strength × source reliability × temporal consistency. **Supporting evidence + contradictory evidence → Confidence**



Python

Core backend,
data processing
& analytics



Playwright

Automated
browser-based
web data
collection



Neo4j

Graph database
for actor
relationships



PyTorch

ML models for
text &
behavioural
analysis



Hugging Face

NLP models and
pretrained
transformers



Next.js

Investigator
dashboard &
frontend



FastAPI

High-
performance
backend APIs



Grafana

Monitoring,
analytics &
system
dashboards



Technical Feasibility

All core components use robust technologies:

- Tor-aware collection
- PostgreSQL + Neo4j
- NLP / stylometry
- Graph-based entity resolution
- Evidence-fusion scoring
- React + FastAPI dashboard

Why feasible: the research architecture maps each component to a concrete implementation technology.



Scalability & Cost Feasibility

- MVP operates with lightweight collectors, asynchronous processing and incremental graph growth.
- High-cost components such as large-scale crawling and blockchain graph analysis are not required initially.
- Collection and analytics can scale horizontally as data volume and investigation requirements increase.



Data Feasibility

We do not require complete dark web coverage. It can incrementally operate on:

- Public/approved onion seeds
- Historical/authorized intelligence
- Controlled synthetic observations

The system stores raw evidence + timestamp + source + provenance, allowing the dataset to grow incrementally. Underground data contains duplicates, reposts, false claims and irrelevant mentions; validation, deduplication, normalization and relevance filtering reduce noise before correlation



Development & Operational Feasibility

The architecture is modular, so development can happen incrementally:

Collection → Evidence Store → Extraction → Graph → Analytics → Attribution → Dashboard

The platform provides decision support rather than autonomous identification, allowing investigators to validate evidence before attribution conclusions are used.

POTENTIAL CHALLENGES & THEIR MITIGATION

 CHALLENGE	 WHY IT'S DIFFICULT	 IMPACT	 MITIGATION STRATEGY
 Incomplete & Volatile Dark-Web Coverage	Onion services frequently go offline, change addresses or disappear.	Gaps in data and missed observations.	• Seed-based discovery + link expansion • Scheduled re-crawling & historical snapshots
 Actor Evasion & Counter-Forensics	Mixers, VPNs, fake personas, AI-generated text, burner wallets and infrastructure hopping.	Weak or misleading signals; harder identity correlation.	• Multi-signal correlation (PGP, crypto, infrastructure, stylometry, behaviour) • Adversarial-aware models
 Malicious Website & Malware Exposure	Suspicious sites may contain malicious downloads, exploit code, deceptive links or ransomware-related content.	Investigator systems and collection infrastructure may be compromised.	• Isolated/sandboxed collection environments • Strict file and content handling with malware scanning
 False Attribution Risk	Shared PGP keys, wallets, servers or coincidental overlaps can mislead correlation.	Wrong conclusions and operational/legal risk.	• Evidence fusion with source reliability weights • Contradictory evidence surfaced explicitly
 Data Quality & Noise	Unstructured content contains spam, scams, duplicates and low-information posts.	Poor extraction quality and higher false positives.	• NLP + rule-based extraction • Deduplication & cross-source validation
 Legal, Ethical & Compliance	Sensitive/illegal content, privacy concerns and jurisdictional restrictions.	Legal exposure and misuse risk.	• Collect only from permitted/authorized sources • Maintain provenance, hashing & audit logs

IMPACT & BENEFITS

IMPACT ON INVESTIGATIVE AGENCIES



Reduced Investigation Effort

Surfaces relationships investigators would otherwise correlate by hand.



Cross-Market Continuity

Connects fragmented actor activity across marketplaces, forums & infrastructure.



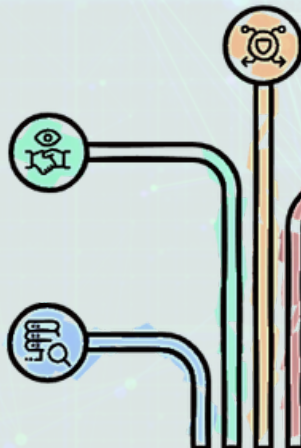
Evidence-Backed Investigation

Preserves raw observations, provenance and timestamps for verification.

IMPACT ON CYBERSECURITY ECOSYSTEM

Resilience Against Actor Churn

Rebranding or migration does not reset investigations.



Better Intelligence Sharing

Structured actor relationships make findings easier to communicate.

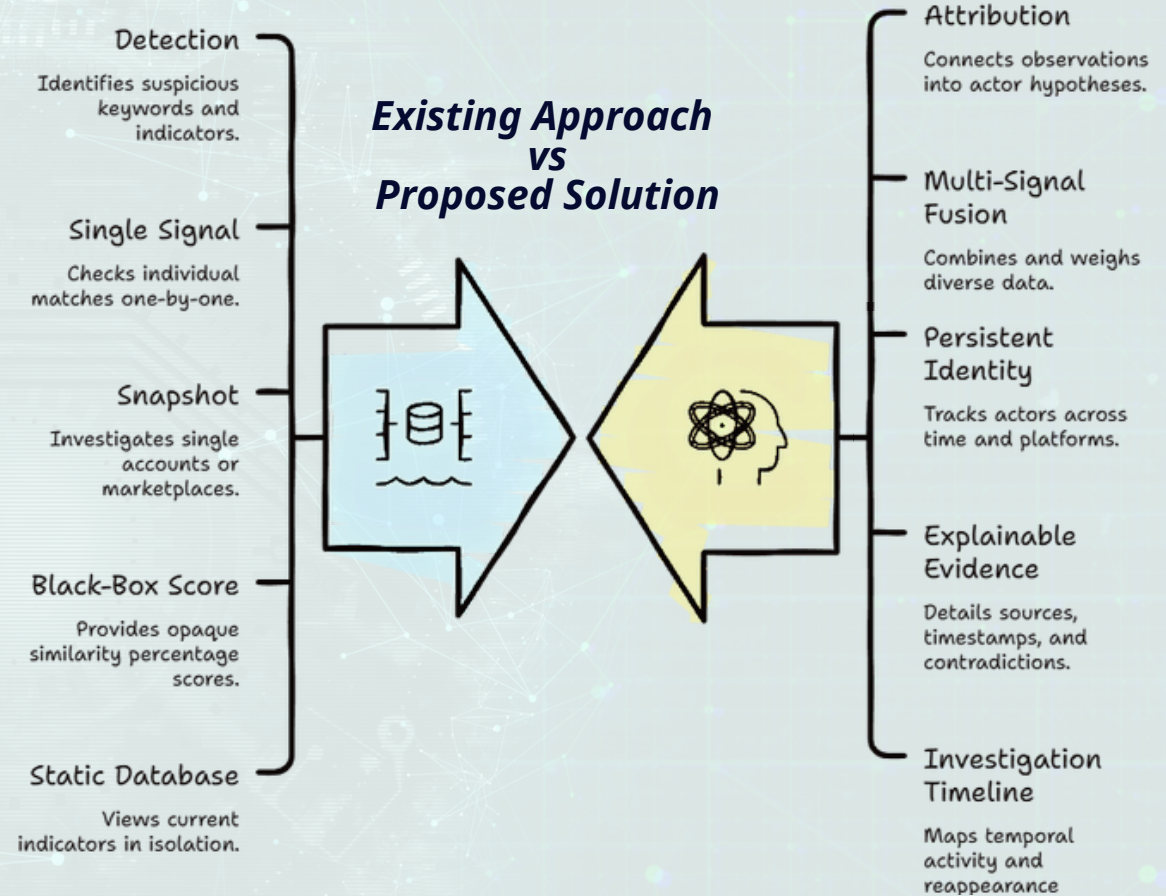
Earlier Threat Discovery

Correlating weak signals can reveal emerging threats before they become conclusive.

Reduced False Attribution

Contradictory evidence remains visible, preventing misidentification.

COMPETITIVE DIFFERENTIATION



BROADER BENEFITS & MEASURABLE VALUE



PUBLIC SAFETY
Supports cybercrime investigations.



RESOURCE EFFICIENCY
Reduces duplicate investigation effort.



KNOWLEDGE RETENTION
Preserves intelligence across cases.



CROSS-CASE INTELLIGENCE
Connects related investigations.



MEASURABLE VALUE
Tracks reuse, coverage and linkages.



SOFTWARE-FIRST
No dedicated field hardware required.

- Dingledine, R., Mathewson, N., & Syverson (2004). Tor: The Second Generation Onion Router. USENIX Security 2004. the foundational Tor paper ; explains exactly why IP layer attribution breaks down [\[Source\]](#)
- Van Wegberg, R., et al. (2018). Plug and Prey? Measuring the Commoditization of Cybercrime via Online Anonymous Markets. USENIX Security 2018. [\[Source\]](#)
- Meiklejohn, S., et al. (2013). A Fistful of Bitcoins: Characterizing Payments Among Men with No Names. origin of the common-input-ownership heuristic; also good for citing mixing/Monero as a known limitation. [\[Source\]](#)
- Ministry of Home Affairs - Indian Cybercrime Coordination Centre (I4C) and National Cybercrime Threat Analytics Unit : The proposed evidence-correlation workflow aligns with India's NCTA Unit objective of analysing fragmented “pieces of puzzles” of cybercrimes to generate actionable threat intelligence. [\[Link\]](#)
- Censys Platform Documentation : Censys provides internet-wide scanning and search data that can support infrastructure investigation and enrichment. [\[Source\]](#)
- Two Hundred Fifty-Fourth Report - Cyber Crime: Ramifications, Protection and Prevention : Provides the Indian institutional and policy context for cybercrime investigation, including the need for coordinated analysis across multiple agencies and specialised cybercrime capabilities. [\[Report\]](#)
- Cyble Vision - Cyber Threat Intelligence and Dark-Web Monitoring - A similar solution
- Positive Technologies - A Look at India Through the Dark Web: What Hackers Are After - 2023-24 [\[Study\]](#)
- DUTA / DUTA-10K — Darknet Usage Text Addresses - Labelled Tor hidden-service data used for validating content classification, relevance filtering and extraction pipelines. [\[Data\]](#)